PROCESS FLOW DIAGRAM (PFD) — ETHANOL-WATER RECOVERY DISTILLATION TRAIN

BASIS: 700.0 kmol/h FEED @ 7.00 mol% ETHANOL | PRODUCT: 82.00 mol% (91.90 wt%) DISTILLATE | RECOVERY: 97.4%

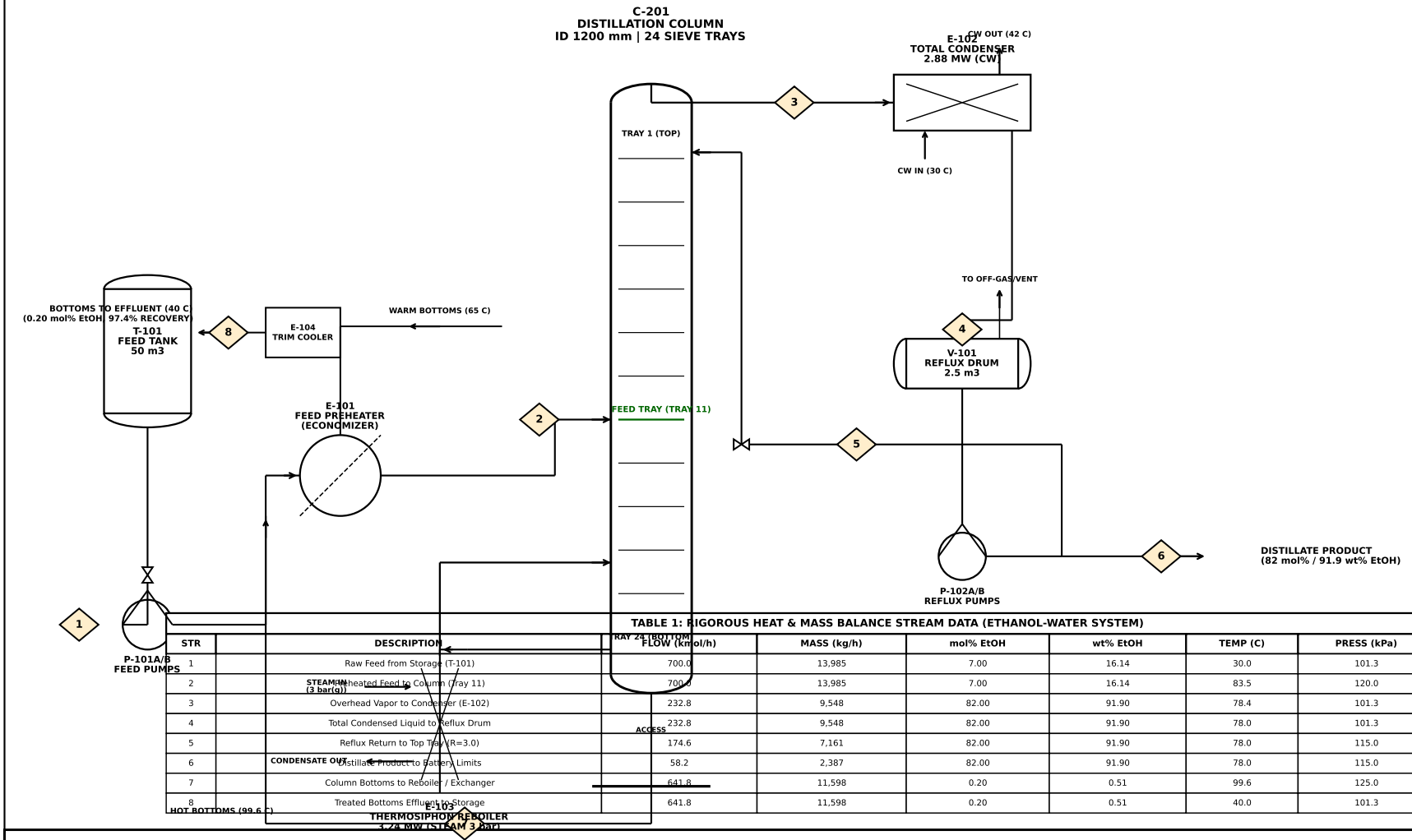

TABLE 1: RIGOROUS HEAT & MASS BALANCE STREAM DATA (ETHANOL-WATER SYSTEM)

STR	DESCRIPTION	FLOW (kmol/h)	MASS (kg/h)	mol% EtOH	wt% EtOH	TEMP (C)	PRESS (kPa)	PHASE
1	Raw Feed from Storage (T-101)	700.0	13,985	7.00	16.14	30.0	101.3	Liquid
2	Preheated Feed to Column (Tray 11)	700.0	13,985	7.00	16.14	83.5	120.0	Bubble Liq (q=1)
3	Overhead Vapor to Condenser (E-102)	232.8	9,548	82.00	91.90	78.4	101.3	Vapor (Dew Pt)
4	Total Condensed Liquid to Reflux Drum	232.8	9,548	82.00	91.90	78.0	101.3	Subcooled Liq
5	Reflux Return to Top Tray (R=3.0)	174.6	7,161	82.00	91.90	78.0	115.0	Liquid
6	Distillate Product to Battery Limits	58.2	2,387	82.00	91.90	78.0	115.0	Liquid Product
7	Column Bottoms to Reboiler / Exchanger	641.8	11,598	0.20	0.51	99.6	125.0	Saturated Liq
8	Treated Bottoms Effluent to Storage	641.8	11,598	0.20	0.51	40.0	101.3	Liquid Effluent

JADAVPUR UNIVERSITY | CHEMICAL ENGINEERING DEPARTMENT | PROCESS EQUIPMENT DESIGN & DRAWING

DRAWING TITLE	LAB / SUBJECT	COURSE / BRANCH	SCALE
PROCESS FLOW DIAGRAM (PFD) & HEAT-MASS BALANCES	PROCESS EQUIPMENT DESIGN & DRAWING	B.E. CHEMICAL ENGINEERING	DIAGRAMMATIC
STUDENT NAME	**ROLL NO.**	**DRAWING NO.**	**DATE**
SOURISH SENAPATI	002310301035	JU/ChE/ED/DWG-01	01 / 09 / 2026

ETHANOL DISTILLATION (C-201) | 700 kmol/h FEED (7 mol% EtOH) | ID 1.20 m, R=3.0 | ASME SEC VIII / TEMA